

CSCI 5352

Network Analysis and Modeling

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (72.1 / 100)	56.4 / 100

56

CURRENT AVG SCORE
across 6 job roles

90

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Examines modern techniques for analyzing and modeling the structure and dynamics of complex networks. Focuses on statistical algorithms and methods, and emphasizes model interpretability and understanding the processes that generate real data. Applications are drawn from computational biology and computational social science. No biological or social science training is required.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	72.1	100.0	+27.9	MEDIUM
Data Scientist	67.1	100.0	+32.9	HIGH
Machine Learning Engineer	62.1	100.0	+37.9	HIGH
ML Engineer	51.7	75.4	+23.7	MEDIUM
AI Engineer	50.3	94.4	+44.1	HIGH
MLOps Engineer	35.2	71.3	+36.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

2 of 6 roles

insights

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

GAP KEYWORDS BY ROLE

Applied Scientist

72.1 → 100.0 (+27.9 pts)

research

literature

problems

models

technical

agents

learning

systems

Data Scientist

67.1 → 100.0 (+32.9 pts)

sets

insights

different

build

problems

Machine Learning Engineer

62.1 → 100.0 (+37.9 pts)

sets

learning

tools

solutions

business

design

development

models

ML Engineer

51.7 → 75.4 (+23.7 pts)

learning

machine

systems

models

build

problems

impact

technical

AI Engineer

50.3 → 94.4 (+44.1 pts)

models

learning

build

software

technical

systems

platform

problems

MLOps Engineer

35.2 → 71.3 (+36.1 pts)

detection

text

systems

learning

machine

image

tools

models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.